



Building a wormhole

In a recent paper, physicists revealed the step-by-step process of creating a wormhole. <https://digit.in/feb20-17>



Our universe in a black hole

One theory suggests that our universe could actually be trapped within one large black hole. <https://digit.in/feb20-18>

An artist's impression of a black hole with a galaxy in the background.

WHITE HOLES

We have heard of black holes, but what are white holes?

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hite holes are theoretical astronomical objects that have the opposite characteristics of

black holes. As against regions of spacetime that are so massive and dense that nothing ever escapes them, white holes are regions in spacetime that just spew out matter and energy. There is no way for anything to enter the white hole from the outside. The pressure from the energetic emissions would keep any object at bay, even though white holes are expected to have mass, and consequently, exert a gravitational influence on all objects.

While formulating the general theory of relativity, Albert Einstein used a set of equations known as the Einstein field equations. First published in 1915, these equations can be used to determine the geometry of spacetime, using mass-energy and momentum. The hyphenation between mass and energy is due to the equivalence between them established by Einstein's most famous equation, $E=mc^2$. The equations are in function similar to how electromagnetic fields can be determined using charge and currents using Maxwell's equations. Now, physicists can fiddle around with the equations and construct all kinds of ridiculous theoretical objects. One such "out there" solution was the black hole, something that remained theoretical for most of the latter half of the 20th century. It is not possible to pinpoint a particular event where the scientific consensus on the existence of black holes went from skeptical to being accepted. However,

most of the dramatic breakthroughs in the field have come within the past five years. In 2015, the LIGO collaboration used gravitational wave astronomy to observe the first collision events between two black holes. In 2019, data from the Event Horizon Telescope was used to publish the first direct image of a black hole. It is now widely accepted that there are thousands of black holes in the universe, including a supermassive black hole at the heart of the Milky Way, known as Sagittarius A*. Scientists believe that most black holes are formed by the collapse of a star, after a star dies and goes nova. Primordial black holes are believed to be formed at the dawn of the universe itself, and supermassive black holes are formed at the time of the creation of the galaxies that they form the hearts of. Two black holes can also merge into a single black hole. From obtuse theoretical objects, black holes have now become very real bodies that can be studied using direct observations from astronomical instruments.

Now, another one of these crazy theoretical objects that can exist because of creative solutions of Einstein's field equations, is a white hole. The first person to make such a suggestion was a Russian cosmologist named Igor Novikov, in 1964. The thing is, the very equations that allow the black holes to exist, also predict the white holes to exist. Imagine observing a black hole in reverse. If that is difficult, just imagine a camera capturing a video recording of a black hole. Matter goes in, and is swallowed up. Play the video in reverse, and you get matter popping out continuously. Einstein's field equations are time invariant, they hold true irrespective of which direction time is moving in. This means that theoretically, white holes exist just because black holes exist. However, at present, most scientists do not think that white holes exist. The main thing here is that white holes can exist, and are consistent with Einstein's theory of general relativity. Another theoretical object that can exist because of Einstein's field equations, is what is called a wormhole.